3.2 MEASURES OF CENTRAL TENDENCY

In statistics, a central tendency or measure of central tendency is a central or typical value of a probability distribution. It is also called a center or location of the distribution. Measures of central tendency are often called averages. An average is a single value which can be taken as a representative of the whole distribution. There are five types of measures of central tendency or averages which are commonly used.

(i) Arithmetic mean or mean or expectation
(ii) Median
(iii) Mode
(iv) Geometric mean
(v) Harmonic mean

1. Mean The mean or average value (μ) of the probability distribution of a discrete random variable X is called as expectation and is denoted by $E(X)$.

$$\mu = E(X) = \sum_{i=1}^{\infty} x_i \; p(x_i) = \sum x \; p(x)$$

where $p(x)$ is the probability mass function of the discrete random variable X. Expectation of any function $\phi(x)$ of a random variable X is given by

$$E[\phi(x)] = \sum_{i=1}^{\infty} \phi(x_i)\, p(x_i) = \sum \phi(x)\, p(x)$$

Some important results on expectation:

(i) $E(X + k) = E(X) + k$
(ii) $E(aX \pm b) = aE(X) \pm b$
(iii) $E(X + Y) = E(X) + E(Y)$ provided $E(X)$ and $E(Y)$ exists.
(iv) $E(XY) = E(X)\, E(Y)$ if X and Y are two independent random variables.

2. Median The median is the point which divides the entire distribution into two equal parts. If X is a random variable, the value of $X = x$ for which the cumulative distribution function $F(x) = \frac{1}{2}$ is called the median of X. For a discrete random variable X, if there exits no x such that $F(x) = \frac{1}{2}$ then the median M of probability distribution is given by

$$M = \frac{1}{2}(x_k + x_{k+1})$$

where $F(x_k) < \frac{1}{2}$ and $F(x_{k+1}) > \frac{1}{2}$ and x_k and x_{k+1} are two consecutive values of X.

3. Mode The mode is the value of discrete random variable X for which the probability is maximum.

4. Geometric Mean The geometric mean G of a random variable X is defined by $\log G = E(\log X)$. The geometric mean of the probability distribution of a discrete random variable X is given by

$$\log G = \sum_{i=1}^{\infty} (\log x_i) p(x_i) = \sum (\log x)\, p(x)$$

where $p(x)$ is the probability mass function of the discrete random variable X.

5. Harmonic Mean The harmonic mean of a random variable X is defined by $\frac{1}{H} = E\left(\frac{1}{X}\right)$. The harmonic mean of the probability distribution of a discrete random variable X is given by

$$\frac{1}{H} = \sum_{i=1}^{\infty} \frac{1}{x_i} p(x_i) = \sum \frac{1}{x} p(x)$$

where $p(x)$ is the probability mass function of the discrete random variable X.

3.3 MEASURES OF DISPERSION

A measure of central tendency is a representative value of the random variable. But it is important to know how the values are clustered around or scattered away from the measure of central tendency. The property of the random variable or its distribution by which its values are clustered around or scattering away from the central value is called dispersion. There are three types of measures of dispersion which are commonly used.

(i) Quartile Deviation
(ii) Mean Deviation
(iii) Standard Deviation

1. Quartile Deviation Quartile deviation or semi-inter quartile range of the probability distribution of a discrete random variable X is given by

$$Q = \frac{1}{2}(Q_3 - Q_1)$$

where Q_1 and Q_3 are the first and third quartiles of the distribution respectively.

2. Mean Deviation Mean deviation of the probability distribution of a discrete random variable X is given by

$$MD = E\{|X - \mu|\}$$

$$= \sum_{i=1}^{\infty} |x_i - \mu| p(x_i)$$

$$= \sum |x - \mu| p(x)$$

where $p(x)$ is the probability mass function of the discrete random variable X.

3. Standard Deviation *Standard deviation* is the positive square root of the arithmetic mean of the squares of the deviations of the given values from their arithmetic mean. It is denoted by the Greek letter σ.

$$\text{SD} = \sigma = \sqrt{\sum_{i=1}^{\infty} x_i^2 \, p(x_i) - \mu^2}$$

$$= \sqrt{E(X^2) - \mu^2}$$

$$= \sqrt{E(X^2) - [E(X)]^2}$$

Variance Variance characterizes the variability in the distributions since two distributions with same mean can still have different dispersion of data about their means. Variance of the probability distribution of a discrete random variable X is given by

$$\begin{aligned}
\text{Var}(X) = \sigma^2 &= E(X-\mu)^2 \\
&= E(X^2 - 2X\mu + \mu^2) \\
&= E(X^2) - E(2X\mu) + E(\mu^2) \\
&= E(X^2) - 2\mu\, E(X) + \mu^2 \qquad [\because E(\text{constant}) = (\text{constant})] \\
&= E(X^2) - 2\mu\mu + \mu^2 \\
&= E(X^2) - \mu^2 \\
&= E(X^2) - [E(X)]^2
\end{aligned}$$

Some important results on variance:

(i) Var $(k) = 0$
(ii) Var $(kX) = k^2$ Var (X)
(iii) Var $(X + k) =$ Var (X)
(iv) Var $(aX + b) = a^2$ Var(X)

Example 1

A random variable X has the following distribution:

X	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Find (i) mean, (ii) variance, and (iii) $P(1 < X < 6)$.

Solution

(i) $\text{Mean} = \mu = \sum xp(x)$

$$= 1\left(\frac{1}{36}\right) + 2\left(\frac{3}{36}\right) + 3\left(\frac{5}{36}\right) + 4\left(\frac{7}{36}\right) + 5\left(\frac{9}{36}\right) + 6\left(\frac{11}{36}\right)$$

$$= \frac{161}{36}$$

$$= 4.47$$

(ii) Variance $= \sigma^2 = \sum x^2 p(x) - \mu^2$

$$= 1\left(\frac{1}{36}\right) + 4\left(\frac{3}{36}\right) + 9\left(\frac{5}{36}\right) + 16\left(\frac{7}{36}\right) + 25\left(\frac{9}{36}\right) + 36\left(\frac{11}{36}\right) - (4.47)^2$$

$$= \frac{791}{36} - 19.98$$

$$= 1.99$$

(iii) $P(1 < X < 6) = P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$

$$= \frac{3}{36} + \frac{5}{36} + \frac{7}{36} + \frac{9}{36}$$

$$= \frac{24}{36}$$

$$= 0.67$$

Example 2

The probability distribution of a random variable X is given below. Find (i) E(X), (ii) Var(X), *(iii) E(2X – 3), and (iv)* Var (2X – 3)

X	–2	–1	0	1	2
$P(X = x)$	0.2	0.1	0.3	0.3	0.1

Solution

(i) $E(X) = \sum x\, p(x)$

$= -2(0.2) - 1(0.1) + 0 + (0.3) + 2(0.1)$

$= 0$

(ii) $\text{Var}(X) = \sum x^2\, p(x) - [E(X)]^2$

$= 4(0.2) + 1(0.1) + 0 + 1(0.3) + 4(0.1) - 0$

$= 1.6$

(iii) $E(2X - 3) = 2E(X) - 3$

$= 2(0) - 3$

$= -3$

(iv) $\text{Var}\,(2X - 3) = (2)^2\,\text{Var}\,(X)$

$= 4(1.6)$

$= 6.4$

Example 3

Mean and standard deviation of a random variable X are 5 and 4 respectively. Find $E(X^2)$ and standard deviation of $(5 - 3X)$.

Solution

$$E(X) = \mu = 5$$
$$\text{SD} = \sigma = 4$$
$$\therefore \quad \text{Var}(X) = \sigma^2 = 16$$
$$\text{Var}(X) = E(X^2) - [E(X)]^2$$
$$16 = E(X^2) - (5)^2$$
$$\therefore \quad E(X^2) = 41$$
$$\text{Var}\,(5 - 3X) = \text{Var}\,(5) - (-3)^2\,\text{Var}\,(X)$$
$$= 0 + 9(16)$$
$$= 144$$
$$\text{SD}\,(5 - 3X) = \sqrt{\text{Var}\,(5 - 3X)}$$
$$= \sqrt{144}$$
$$= 12$$

Example 4

A machine produces an average of 500 items during the first week of the month and on average of 400 items during the last week of the month, the probability for these being 0.68 and 0.32 respectively. Determine the expected value of the production. **[Summer 2015]**

Solution

Let X be the random variable which denotes the items produced by the machine. The probability distribution is

X	500	400
$P(X = x)$	0.68	0.32

Expected value of the production $E(X) = \sum x\, p(x)$

$$= 500(0.68) + 400(0.32)$$
$$= 468$$

Example 5

The monthly demand for Allwyn watches is known to have the following probability distribution:

Demand (x)	1	2	3	4	5	6	7	8
Probability $p(x)$	0.08	0.12	0.19	0.24	0.16	0.10	0.07	0.04

Find the expected demand for watches. Also, compute the variance.

Solution

$$\begin{aligned}E(X) &= \Sigma x\, p(x)\\ &= 1(0.08)+2(0.12)+3(0.19)+4(0.24)+5(0.16)\\ &\quad +6(0.10)+7(0.07)+8(0.04)\\ &= 4.06\\ \text{Var}(X) &= E(X^2)-[E(X)]^2\\ &= \Sigma\, x^2\; p(x)-[E(X)]^2\\ &= 1(0.08)+4(0.12)+9(0.19)+16(0.24)+25(0.16)\\ &\quad +36(0.10)+49(0.07)+64(0.04)-(4.06)^2\\ &= 19.7-16.48\\ &= 3.21\end{aligned}$$

Example 6

A discrete random variable has the probability mass function given below:

X	−2	−1	0	1	2	3
$P(X=x)$	0.2	k	0.1	$2k$	0.1	$2k$

Find k, mean, and variance.

Solution

Since $P(X=x)$ is a probability mass function,

$$\sum P(X=x)=1$$

$$\begin{aligned}0.2+k+0.1+2k+0.1+2k &= 1\\ 5k+0.4 &= 1\\ 5k &= 0.6\\ k &= \frac{0.6}{5}=\frac{3}{25}\end{aligned}$$

Hence, the probability distribution is

X	−2	−1	0	1	2	3
$P(X=x)$	$\frac{2}{10}$	$\frac{3}{25}$	$\frac{1}{10}$	$\frac{6}{25}$	$\frac{1}{10}$	$\frac{6}{25}$

$$\text{Mean} = E(X) = \sum x\,p(x)$$
$$= (-2)\left(\frac{2}{10}\right)+(-1)\left(\frac{3}{25}\right)+0+1\left(\frac{6}{25}\right)+2\left(\frac{1}{10}\right)+3\left(\frac{6}{25}\right)$$
$$= \frac{6}{25}$$

$$\text{Variance} = \text{Var}(X) = E(X^2)-[E(X)]^2$$
$$= \sum x^2\,p(x)-[E(X)]^2$$
$$= 4\left(\frac{2}{10}\right)+1\left(\frac{3}{25}\right)+0+1\left(\frac{6}{25}\right)+4\left(\frac{1}{10}\right)+9\left(\frac{6}{25}\right)-\left(\frac{6}{25}\right)^2$$
$$= \frac{73}{250}-\frac{36}{625}$$
$$= \frac{293}{625}$$

Example 7

A random variable X has the following probability function:

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

(i) *Determine k.* (ii) *Evaluate* $P(X < 6)$, $P(X \geq 6)$, $P(0 < X < 5)$ and $P(0 \leq X \leq 4)$. (iii) *Determine the distribution function of X. (iv) Find the mean. (v) Find the variance.*

Solution

(i) Since $p(x)$ is a probability mass function,

$$\sum p(x) = 1$$
$$0+k+2k+2k+3k+k^2+2k^2+7k^2+k = 1$$
$$10k^2+9k-1 = 1$$
$$(10k-1)(k+1) = 0$$
$$k = \frac{1}{10} \text{ or } k = -1$$
$$k = \frac{1}{10} = 0.1 \quad [\because \; p(x) \geq 0, k \neq -1]$$

Hence, the probability function is

X	0	1	2	3	4	5	6	7
$P(X=x)$	0	0.1	0.2	0.2	0.3	0.01	0.02	0.17

(ii) $P(X<6) = P(X=0)+P(X=1)+P(X=2)+P(X=3)+P(X=4)+P(X=5)$

$= 0+0.1+0.2+0.2+0.3+0.01$

$= 0.81$

$P(X \geq 6) = 1-P(X<6)$

$= 1-0.81$

$= 0.19$

$P(0<X<5) = P(X=1)+P(X=2)+P(X=3)+P(X=4)$

$= 0.1+0.2+0.2+0.3$

$= 0.8$

$P(0 \leq X \leq 4) = P(X=0)+P(X=1)+P(X=2)+P(X=3)+P(X=4)$

$= 0+0.1+0.2+0.2+0.3$

$= 0.8$

(iii) Distribution function of X

x	$p(x)$	$F(x)$
0	0	0
1	0.1	0.1
2	0.2	0.3
3	0.2	0.5
4	0.3	0.8
5	0.01	0.81
6	0.02	0.83
7	0.17	1

(iv) $\mu = \sum xp(x)$

$= 0+1(0.1)+2(0.2)+3(0.2)+4(0.3)+5(0.01)+6(0.02)+7(0.17)$

$= 3.66$

(v) $\text{Var}(X) = \sigma^2 = \sum x^2\, p(x) - \mu^2$

$= 0+1(0.1)+4(0.2)+9(0.2)+16(0.3)+25(0.01)+36(0.02)$

$+49(0.17)-(3.66)^2$

$= 3.4044$

Example 8

A fair dice is tossed. Let the random variable X denote the twice the number appearing on the dice. Write the probability distribution of X. Calculate mean and variance.

Solution

Let X be the random variable which denotes twice the number appearing on the dice.

(i) Probability distribution of X

x	2	4	6	8	10	12
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

(ii) $\text{Mean} = \mu = \sum xp(x)$

$$= 2\left(\frac{1}{6}\right)+4\left(\frac{1}{6}\right)+6\left(\frac{1}{6}\right)+8\left(\frac{1}{6}\right)+10\left(\frac{1}{6}\right)+12\left(\frac{1}{6}\right)$$

$$= 7$$

(iii) $\text{Variance} = \sigma^2 = \sum x^2 p(x) - \mu^2$

$$= 4\left(\frac{1}{6}\right)+16\left(\frac{1}{6}\right)+36\left(\frac{1}{6}\right)+64\left(\frac{1}{6}\right)+100\left(\frac{1}{6}\right)+144\left(\frac{1}{6}\right)-(7)^2$$

$$= 11.67$$

Example 9

Two unbiased dice are thrown at random. Find the probability distribution of the sum of the numbers on them. Also, find mean and variance.

Solution

Let X be the random variable which denotes the sum of the numbers on two unbiased dice. The random variable X can take values 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. The probability distribution is

X	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$\text{Mean} = \mu = \sum x\, p(x)$

$$= 2\left(\frac{1}{36}\right)+3\left(\frac{2}{36}\right)+4\left(\frac{3}{36}\right)+5\left(\frac{4}{36}\right)+6\left(\frac{5}{36}\right)+7\left(\frac{6}{36}\right)+8\left(\frac{5}{36}\right)$$
$$+9\left(\frac{4}{36}\right)+10\left(\frac{3}{36}\right)+11\left(\frac{2}{36}\right)+12\left(\frac{1}{36}\right)$$
$$= \frac{252}{36}$$
$$= 7$$

$$\text{Variance} = \sigma^2 = \sum x^2 p(x) - \mu^2$$
$$= 4\left(\frac{1}{36}\right)+9\left(\frac{2}{36}\right)+16\left(\frac{3}{36}\right)+25\left(\frac{4}{36}\right)+36\left(\frac{5}{36}\right)$$
$$+49\left(\frac{6}{36}\right)+64\left(\frac{5}{36}\right)+81\left(\frac{4}{36}\right)+100\left(\frac{3}{36}\right)$$
$$+121\left(\frac{2}{36}\right)+144\left(\frac{1}{36}\right)-(7)^2$$
$$= \frac{1974}{36} - 49$$
$$= 5.83$$

Example 10

A sample of 3 items is selected at random from a box containing 10 items of which 4 are defective. Find the expected number of defective items.

Solution

Let X be the random variable which denotes the defective items.

Total number of items = 10

Number of good items = 6

Number of defective items = 4

$$P(X=0) = P(\text{no defective item}) = \frac{{}^6C_3}{{}^{10}C_3} = \frac{1}{6}$$

$$P(X=1) = P(\text{one defective item}) = \frac{{}^6C_2\ {}^4C_1}{{}^{10}C_3} = \frac{1}{2}$$

$$P(X=2) = P(\text{two defective items}) = \frac{{}^6C_1\ {}^4C_2}{{}^{10}C_3} = \frac{3}{10}$$

$$P(X=3) = P(\text{three defective items}) = \frac{{}^4C_3}{{}^{10}C_3} = \frac{1}{30}$$

Hence, the probability distribution is

X	0	1	2	3
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

Expected number of defective items $= E(X) = \sum x\,p(x)$

$$= 0 + 1\left(\frac{1}{2}\right) + 2\left(\frac{3}{10}\right) + 3\left(\frac{1}{30}\right)$$

$$= 1.2$$

Example 11

A player tosses two fair coins. He wins ₹ 100 *if a head appears and* ₹ 200 *if two heads appear. On the other hand, he loses* ₹ 500 *if no head appears. Determine the expected value of the game. Is the game favourable to the players?*

Solution

Let X be the random variable which denotes the number of heads appearing in tosses of two fair coins.

$$S = \{HH, HT, TH, TT\}$$

$$p(x_1) = P(X=0) = P(\text{no heads}) = \frac{1}{4}$$

$$p(x_2) = P(X=1) = P(\text{one head}) = \frac{2}{4} = \frac{1}{2}$$

$$p(x_3) = P(X=2) = P(\text{two heads}) = \frac{1}{4}$$

Amount to be lost if no head appears $= x_1 = -$ ₹ 500

Amount to be won if one head appears $= x_2 =$ ₹ 100

Amount to be won if two heads appear $= x_3 =$ ₹ 200

Expected value of the game $= \mu = \sum x\,p(x)$

$$= x_1\,p(x_1) + x_2\,p(x_2) + x_3\,p(x_3)$$

$$= -500\left(\frac{1}{4}\right) + 100\left(\frac{1}{2}\right) + 200\left(\frac{1}{4}\right)$$

$$= ₹ -25$$

Hence, the game is not favourable to the player.

Example 12

Amit plays a game of tossing a dice. If a number less than 3 appears, he gets ₹ a, otherwise he has to pay ₹ 10. If the game is fair, find a.

Solution

Let X be the random variable which denotes tossing of a dice.

Probability of getting a number less than 3, i.e., 1 or 2 = $p(x_1) = \frac{2}{6} = \frac{1}{3}$

Probability of getting number more than or equal to 3, i.e., 3, 4, 5, or 6 = $p(x_2) = \frac{4}{6} = \frac{2}{3}$

Amount to be received for number less than 3 = x_1 = ₹ a

Amount to be paid for numbers more than or equal to 3 = x_2 = ₹ -10

$$E(X) = \sum x\, p(x)$$

$$= x_1\, p(x_1) + x_2\, p(x_2)$$

$$= a\left(\frac{1}{3}\right) + (-10)\left(\frac{2}{3}\right)$$

$$= \frac{a}{3} - \frac{20}{3}$$

For a pair game, $E(x) = 0$.

$$\frac{a}{3} - \frac{20}{3} = 0$$

$$a = 20$$

Example 13

A man draws 2 balls from a bag containing 3 white and 5 black balls. If he is to receive ₹ 14 for every white ball which he draws and ₹ 7 for every black ball, what is his expectation?

Solution

Let X be the random variable which denotes the balls drawn from a bag. 2 balls drawn may be either (i) both white, or (ii) both black, or (iii) one white and one black.

Probability of drawing 2 white balls = $p(x_1) = \frac{{}^3C_2}{{}^8C_2} = \frac{3}{28}$

Probability of drawing 2 black balls = $p(x_2) = \frac{{}^5C_2}{{}^8C_2} = \frac{10}{28}$

Probability of drawing 1 white and 1 black ball = $p(x_3) = \frac{{}^3C_1 \, {}^5C_1}{{}^8C_2} = \frac{15}{28}$

Amount to be received for 2 white balls = x_1 = ₹ 14 × 2 = ₹ 28
Amount to be received for 2 black balls = x_2 = ₹ 7 × 2 = ₹ 14
Amount to be received for 1 white and 1 black ball = x_3 = ₹ 14 + ₹ 7 = ₹ 21

Expectation = $E(X) = \sum x\,p(x)$

$$= x_1\,p(x_1) + x_2\,p(x_2) + x_3\,p(x_3)$$

$$= 28\left(\frac{3}{28}\right) + 14\left(\frac{10}{28}\right) + 21\left(\frac{15}{28}\right)$$

$$= ₹\,19.25$$

Example 14

The probability that there is at least one error in an account statement prepared by A is 0.2 *and for B and C, they are* 0.25 *and* 0.4 *respectively. A, B, and C prepared* 10, 16, *and* 20 *statements respectively. Find the expected number of correct statements in all.*

Solution

Let $p(x_1)$, $p(x_2)$ and $p(x_3)$ be the probabilities of the events that there is no error in the account statements prepared by *A*, *B*, and *C* respectively.

$p(x_1) = 1 - $ (Probability of at least one error in the account statement prepared by *A*)

$= 1 - 0.2$

$= 0.8$

Similarly, $p(x_2) = 1 - 0.25 = 0.75$

$p(x_3) = 1 - 0.4 = 0.6$

Also, $x_1 = 10, \quad x_2 = 16, \quad x_3 = 20$

Expected number of correct statements = $E(X) = \sum x\,p(x)$

$$= x_1\,p(x_1) + x_2\,p(x_2) + x_3\,p(x_3)$$

$$= 10(0.8) + 16(0.75) + 20(0.6)$$

$$= 32$$

Example 15

A man has the choice of running either a hot-snack stall or an ice-cream stall at a seaside resort during the summer season. If it is a fairly cool

summer, he should make ₹ 5000 by running the hot-snack stall, but if the summer is quite hot, he can only expect to make ₹ 1000. On the other hand, if he operates the ice-cream stall, his profit is estimated at ₹ 6500, if the summer is hot, but only ₹ 1000 if it is cool. There is a 40 percent chance of the summer being hot. Should he opt for running the hot-snack stall or the ice-cream stall?

Solution

Let X and Y be the random variables which denote the income from the hot-snack and ice-cream stalls respectively.

Probability of hot summer $= p_1 = 40\% = 0.4$

Probability of cool summer $= p_2 = 1 - p_1 = 1 - 0.4 = 0.6$

$$x_1 = 1000, \quad x_2 = 5000, \quad y_1 = 6500, \quad y_2 = 1000$$

Expected income from hot-snack stall $= E(X)$

$$= x_1 p_1 + x_2 p_2$$
$$= 1000(0.4) + 5000(0.6)$$
$$= ₹\ 3400$$

Expected income from ice-cream stall $= E(Y)$

$$= y_1 p_1 + y_2 p_2$$
$$= 6500\ (0.4) + 1000(0.6)$$
$$= ₹\ 3200$$

Hence, he should opt for running the hot-snack stall.

3.4 MOMENTS

Moment is the arithmetic mean of the various powers of the deviations of items from their assumed mean or actual mean. If the deviations of the items are taken from the arithmetic mean of the distribution, it is known as *central moment*. If the mean of the first power of deviations are taken, the first moment about the mean is obtained and is denoted by μ_1. The mean of the second power of the deviations gives the second moment about the mean and is denoted by μ_2. Similarly, the mean of the cubes of deviations gives third moment about the mean and is denoted by μ_3. The mean of the fourth power of the deviations from the mean gives the fourth moment about the mean and is denoted by μ_4. Thus, the mean of the r^{th} power of deviations gives the r^{th} moment about mean or r^{th} central moment and is denoted by μ_r.

3.4.1 Central Moments or Moments about Actual Mean

The moments about the mean value $\mu = E(X)$ are called central moments and denoted by μ_r.

$$\mu_r = E\{(x-\mu)^r\}$$

$$= \sum_{i=1}^{\infty}(x_i-\mu)^r\, p(x_i)$$

$$= \sum(x-\mu)^r p(x)$$

If frequency distribution is given and $n = \Sigma f$, then $p(x_i) = \dfrac{\sum f_i}{N}$

$$\mu_r = \frac{\sum_{i=1}^{\infty} f_i(x_i-\mu)^r}{N}$$

3.4.2 Properties of Central Moments

(i) The first moment about the mean is always zero, i.e., $\mu_1 = 0$.

(ii) The second moment about the mean measures variance, i.e.,

$$\mu_2 = \sigma^2 \text{ or } SD = \sigma = \pm\sqrt{\mu_2}$$

(iii) The third moment about the mean measures skewness.

If $\mu_3 > 0$, the distribution is positively skewed.
If $\mu_3 < 0$, the distribution is negatively skewed.
If $\mu_3 = 0$, the distribution is symmetrical.

$$\text{Skewness } \beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

(iv) The fourth moment about the mean measures kurtosis. It gives information on the peakedness or height of the peak of a frequency distribution, i.e., whether it is more peaked or more flat topped than a normal curve.

$$\text{Kurtosis } \beta_2 = \frac{\mu_4}{\mu_2^2}$$

(v) In a symmetric distribution, all odd moments are zero, i.e., $\mu_1 = \mu_3 = \mu_5 = \cdots = \mu_{2r+1} = 0$.

3.4.3 Raw Moments or Moments about Arbitrary Origin

When the actual mean of a distribution is a fraction, it is tedious to calculate central moments. In such cases, moments about an arbitrary origin 'a' is calculated and then these moments are converted into the moments about actual mean. The moments about the arbitrary origin are known as raw moments and are denoted by μ_r'. Thus, μ_1' denotes the first moment about an arbitrary origin, μ_2' denotes the second moment about an arbitrary origin and so on.

$$\begin{aligned}\mu_r' &= E\{(X-a)^r\} \\ &= \sum_{i=1}^{\infty}(x_i - a)^r p(x_i) \\ &= \sum (x-a)^r p(x)\end{aligned}$$

If frequency distribution is given and $n = \Sigma f$, then $p(x_i) = \dfrac{\sum f_i}{N}$

$$\mu_r' = \frac{\sum_{i=1}^{\infty} f_i (x_i - a)^r}{N}$$

When $a = 0$, μ_r' is called r^{th} order simple moments.

$$\begin{aligned}\mu_r' &= E\{X^r\} \\ &= \sum_{i=1}^{\infty} x_i^r p(x_i)\end{aligned}$$

$$= \sum x^r p(x)$$

$$= \frac{\sum fx^r}{n}$$

3.4.4 Relation between Central Moments and Raw Moments

The moments about the actual mean, i.e., central moments and moments about the arbitrary origin, i.e., raw moments are related with each other by the following equations:

First central moment $\mu_1 = \mu_1' - \mu_1' = 0$

Second central moment $\mu_2 = \mu_2' - (\mu_1')^2$

Third central moment $\mu_3 = \mu_3' - 3\mu_2' \mu_1' + 2(\mu_1')^3$

Fourth central moment $\mu_4 = \mu_4' - 4\mu_3' \mu_1' + 6\mu_2' (\mu_1')^2 - 3(\mu_1')^4$

Similarly, the raw moments can be expressed in terms of central moments.

First raw moment $\mu_1' = \mu - a$

Second raw moment $\mu_2' = \mu_2 + (\mu_1')^2$

Third raw moment $\mu_3' = \mu_3 + 3\mu_2 \mu_1' + (\mu_1')^3$

Fourth raw moment $\mu_4' = \mu_4 + 4\mu_3 \mu_1' + 6\mu_2(\mu_1')^2 + (\mu_1')^4$

Example 1

Calculate the first four moments from the following data:

x	0	1	2	3	4	5	6	7	8
f	5	10	15	20	25	20	15	10	5

Also, calculate the values of β_1 and β_2.

Solution

$$N = \sum f = 125$$

$$\bar{x} = \frac{\sum fx}{N} = \frac{500}{125} = 4$$

x	f	fx	$x-\mu$	$f(x-\mu)$	$f(x-\mu)^2$	$f(x-\mu)^3$	$f(x-\mu)^4$
0	5	0	–4	20	80	–320	1280
1	10	10	–3	–30	90	–270	810
2	15	30	–2	–30	60	–120	240
3	20	60	–1	–20	20	–20	20
4	25	100	0	0	0	0	0
5	20	100	1	20	20	20	20
6	15	90	2	30	60	120	240
7	10	70	3	30	90	270	810
8	5	40	4	20	80	320	1280
	Σf = 125	Σfx = 500		$\Sigma f(x-\mu)$ = 0	$\Sigma f(x-\mu)^2$ = 500	$\Sigma f(x-\mu)^3 = 0$	$\Sigma f(x-\mu)^4$ = 4700

Moments about the actual mean:

$$\mu_1 = \frac{\sum f(x-\mu)}{N} = \frac{0}{125} = 0$$

$$\mu_2 = \frac{\sum f(x-\mu)^2}{N} = \frac{500}{125} = 4$$

$$\mu_3 = \frac{\sum f(x-\mu)^3}{N} = \frac{0}{125} = 0$$

$$\mu_4 = \frac{\sum f(x-\mu)^4}{N} = \frac{4700}{125} = 37.6$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{0}{64} = 0$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{37.6}{16} = 2.35$$

Example 2

Calculate the first four moments of the following distribution about the mean:

x	0	1	2	3	4	5	6	7	8
f	1	8	28	56	70	56	28	8	1

Also, evaluate β_1 and β_2.

Solution

Let $a = 4$ be the arbitrary origin.

x	f	$x-a$	$f(x-a)$	$f(x-a)^2$	$f(x-a)^3$	$f(x-a)^4$
0	1	–4	–4	16	–64	256
1	8	–3	–24	72	–216	648
2	28	–2	–56	112	–224	448
3	56	–1	–56	56	–56	56
4	70	0	0	0	0	0
5	56	1	56	56	56	56
6	28	2	56	112	224	448
7	8	3	24	72	216	648
8	1	4	4	16	64	256
	$\sum f = 256$		$\sum f(x-a) = 0$	$\sum f(x-a)^2 = 512$	$\sum f(x-a)^3 = 0$	$\sum f(x-a)^4 = 2816$

$$N = \sum f = 256$$

Moments about the arbitrary origin:

$$\mu_1' = \frac{\sum f(x-a)}{N} = \frac{0}{256} = 0$$

$$\mu_2' = \frac{\sum f(x-a)^2}{N} = \frac{512}{256} = 2$$

$$\mu_3' = \frac{\sum f(x-a)^3}{N} = \frac{0}{256} = 0$$

$$\mu_4' = \frac{\sum f(x-a)^4}{N} = \frac{2816}{256} = 11$$

Moments about the actual mean:

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$
$$= 2 - 0$$
$$= 2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3$$
$$= 0 - 3(2)(0) + 2(0)^3$$
$$= 0$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$$
$$= 11 - 4(0)(0) + 6(2)(0)^2 - 3(0)^4$$
$$= 11$$
$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0$$
$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{11}{(2)^2} = 2.75$$

Example 3

The first four moments of distribution about $x = 2$ are 1, 2.5, 5.5, *and* 16. *Calculate the four moments about μ.*

Solution

$$\mu_1' = 1, \quad \mu_2' = 2.5, \quad \mu_3' = 5.5, \quad \mu_4' = 16$$

Moments about the mean:

$$\mu_1 = 0$$
$$\mu_2 = \mu_2' - (\mu_1')^2$$
$$= 2.5 - (1)^2$$
$$= 1.5$$
$$\mu_3 = \mu_3' - 3\mu_2\mu_1' + 2(\mu_1')^3$$
$$= 5.5 - 3(2.5)(1) + 2(1)^3$$
$$= 0$$
$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$$
$$= 16 - 4(5.5)(1) + 6(2.5)(1)^2 - 3(1)^4$$
$$= 6$$

Example 4

The first three moments of a distribution about the value 2 *of the variables are* 1, 16, *and* –40. *Show that the mean* = 3, *variance* = 15 *and* $\mu_3 = -86$.

Solution

$$a = 2, \quad \mu_1' = 1, \quad \mu_2' = 16, \quad \mu_2' = 16, \quad \mu_3' = -40$$

$$\mu_1' = \mu - a$$
$$1 = \mu - 2$$
$$\therefore \quad \mu = 3$$

Mean = 3

$$\mu_2 = \mu_2' - (\mu_1')^2$$
$$= 16 - (1)^2$$
$$= 15$$

Variance = $\mu_2 = 15$

$$\mu_3 = \mu_3' - 3\mu_2'\,\mu_1' + 2(\mu_1')^3$$
$$= -40 - 3(16)(1) + 2(1)^3$$
$$= -86$$

3.5 SKEWNESS

Skewness is a measure that refers to the extent of symmetry or asymmetry in a distribution. A distribution is said to be *symmetrical* when its mean, median, and mode are equal, and the frequencies are symmetrically distributed about the mean. A symmetrical distribution when plotted on a graph will give a perfectly bell-shaped curve which is known as a *normal curve* (Fig. 3.1).

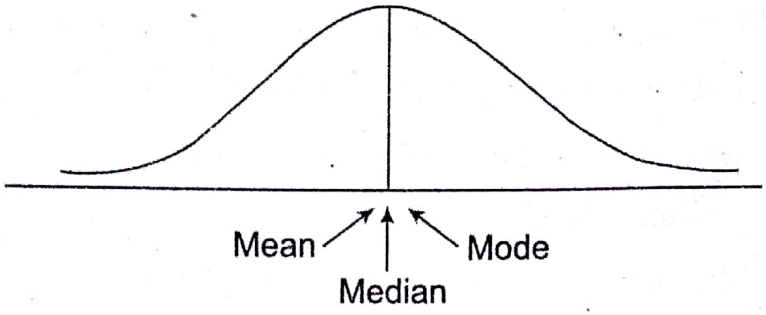
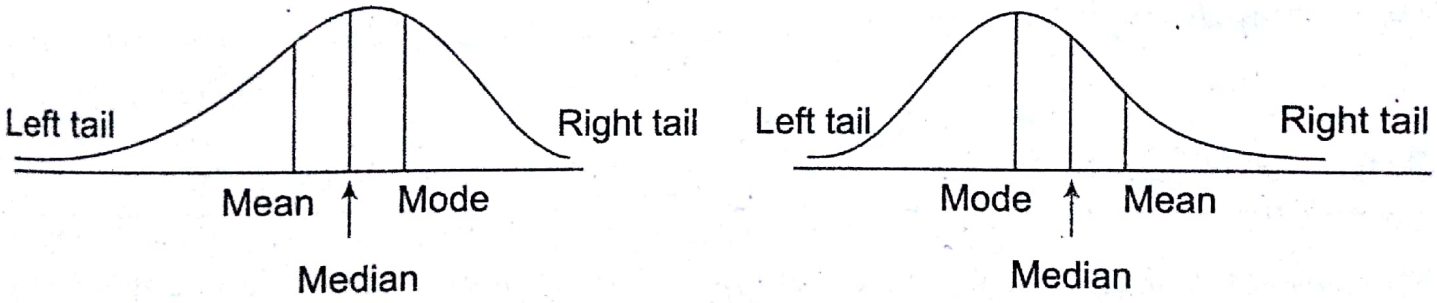


Fig. 3.1

A distribution is said to be asymmetrical or skewed when the mean, median, and mode are not equal, i.e., the mean median, and mode do not coincide. If the curve has a longer tail towards the left, it is said to be a negatively skewed distribution (Fig. 3.2*a*). If the curve has a longer tail towards the right, it is said to be positively skewed (Fig. 3.2*b*).



(a) Negatively skewed distribution (b) Positively skewed distribution

Fig. 3.2

Skewness gives an idea of the nature and degree of concentration of observations about the mean.

3.5.1 Measures of Skewness

A measure of skewness gives the extent and direction of skewness of a distribution. These measures can be absolute or relative. The absolute measures are also known as measures of skewness.

Absolute skewness = Mean – Mode

If the value of the mean is greater than the mode, the skewness will be positive and if the value of the mean is less than the mode, the skewness will be negative.

The relative measures of skewness is called the *coefficient of skewness.*

3.5.2 Karl Pearson's Coefficient of Skewness

Karl Pearson's coefficient of skewness denoted by S_k, is given by

$$S_k = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

$$= \frac{\text{Mean} - \text{Mode}}{\sigma}$$

When the mode is ill-defined and the distribution is moderately skewed, the averages have the following relationship:

$$\text{Mode} = 3\,\text{Median} - 2\,\text{Mean}$$

$$S_k = \frac{\text{Mean} - (3\,\text{Median} - 2\,\text{Mean})}{\text{Standard Deviation}}$$

$$= \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

$$= \frac{3(\text{Mean} - \text{Median})}{\sigma}$$

The coefficient of skewness usually lies between –1 and 1.

For a positively skewed distribution, $S_k > 0$.

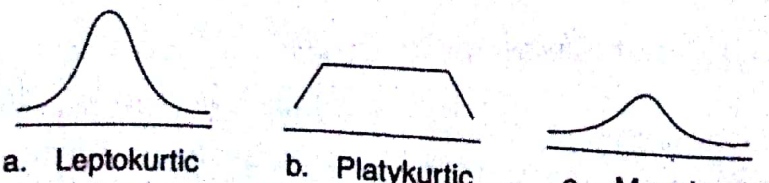
For a negatively skewed distribution, $S_k < 0$.

For a symmetrical distribution, $S_k = 0$.

3.6 KURTOSIS

Measures of central tendency, dispersion and skewness of a random variable cannot give a complete idea about the probability distribution. In order to analyse the probability distribution completely, another characteristic, Kurtosis is required. Kurtosis means the convexity of the probability curve of the distribution. It measures the degree of peakedness of distribution and is given by

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}$$



a. Leptokurtic b. Platykurtic c. Mesokurtic

Fig. 3.3

The curves with $\beta_2 > 3$ is called Leptokurtic and those with $\beta_2 < 3$ are called platykurtic. The normal curve for which $\beta_2 = 3$ is called Mesokurtic.

As $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$ and $\beta_2 = \frac{\mu_4}{\mu_2^2}$ determine the shape of the probability curve, these are called Pearson's shape coefficients.

Example 1

From the marks scored by 100 *students in Section A and* 100 *students in Section B of a class, the following measures were obtained:*

Section *A*	$\mu_A = 55$	$\sigma_A = 15.4$	Mode = 58.72
Section *B*	$\mu_A = 53$	$\sigma_B = 15.4$	Mode = 48.83

Determine which distribution of marks is more skewed.

Solution

$$S_{k_A} = \frac{\text{Mean} - \text{Mode}}{\sigma_A} = \frac{55 - 58.72}{15.4} = -0.24$$

$$S_{k_B} = \frac{\text{Mean} - \text{Mode}}{\sigma_B} = \frac{53 - 48.83}{15.4} = 0.27$$

$|0.27| > |-0.24|$

Hence, the distribution of marks of Section *B* is more skewed.

Example 2

For a group of 10 *items,* $\sum x = 452$, $\sum x^2 = 24270$, *and mode* = 43.7. *Find Karl Pearson's coefficient of skewness.*

Solution

$n = 10$, $\sum x = 452$, $\sum x^2 = 24270$, mode = 43.7

$$\mu = \frac{\Sigma x}{n} = \frac{452}{10} = 45.2$$

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

$$= \sqrt{\frac{24270}{10} - \left(\frac{452}{10}\right)^2}$$

$$= 19.59$$

$$S_k = \frac{\text{Mean} - \text{Mode}}{\sigma}$$
$$= \frac{45.2 - 43.7}{19.59}$$
$$= 0.077$$

Example 3

In a distribution, the mean = 65, median = 70, coefficient of skewness = –0.6. Find the mode and coefficient of variation.

Solution

$$\mu = 65, \quad \text{Median} = 70, \quad S_k = -0.6$$

$$\text{Mode} = 3\,\text{Median} - 2\,\text{Mean} = 3(70) - 2(65) = 80$$

$$S_k = \frac{\text{Mean} - \text{Mode}}{\sigma}$$

$$-0.6 = \frac{65 - 80}{\sigma}$$

$$\therefore \quad \sigma = 25$$

$$\text{CV} = \frac{\sigma}{\bar{x}} \times 100 = \frac{25}{65} \times 100 = 38.64\%$$

Example 4

The following information was obtained from the records of a factory relating to wages:

Arithmetic mean = ₹ 56.8, Median = ₹ 59.5, *Standard deviation* = ₹ 12.4

Give the information about the distribution of wages.

Solution

$$\mu = 56.8, \quad \text{Median} = 59.5, \quad \sigma = 12.4$$

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\sigma} = \frac{3(56.8 - 59.5)}{12.4} = -0.65$$

$$\text{Mode} = 3\,\text{Median} - 2\,\text{Mean} = 3(59.5) - 2(56.8) = 64.9$$

Hence, the maximum wages is ₹ 64.9.

There is a negative skewness in wages.

Example 5

For a moderately skewed distribution of retail price for men's shoes, it is found that the mean price is ₹ 20 and the median price is ₹ 17. If the coefficient of variation is 20%, find the Pearson's coefficient of skewness.

Solution

$$\mu = 20, \quad \text{Median} = 17, \quad \text{CV} = 20\%$$

$$\text{CV} = \frac{\sigma}{\bar{x}} \times 100$$

$$20 = \frac{\sigma}{20} \times 100$$

$$\therefore \quad \sigma = 4$$

$$S_k = \frac{3(\text{Mean} - \text{Median})}{\sigma} = \frac{3(20-17)}{4} = 2.25$$

Example 6

Find the mean, SD, quartiles, median and Karl Pearson's coefficient of skewness for the following probability distribution:

$X = x$	1	2	3	4	5	6	7	8
$p(x)$	0.008	0.032	0.142	0.216	0.240	0.206	0.143	0.013

Solution

(i) $\text{Mean} = \mu = \sum x\, p(x)$

$$= 1(0.008) + 2(0.032) + 3(0.142) + 4(0.216) + 5(0.240) + 6(0.206) + 7(0.143) + 8(0.013)$$

$$= 4.903$$

(ii) $\text{Var}(X) = \sigma^2 = \sum x^2 p(x) - \mu^2$

$$= 1(0.008) + 4(0.032) + 9(0.142) + 16(0.216) + 25(0.240) + 36(0.206) + 49(0.143) + 64(0.013) - (4.903)^2$$

$$= 2.086$$

$$SD = \sqrt{\text{Var}(X)} = \sqrt{2.086} = 1.444$$

(iii) $F(3) = 0.008 + 0.032 + 0.142 = 0.182 < 0.25$

$F(4) = 0.008 + 0.032 + 0.142 + 0.216 = 0.398 > 0.25$

$$Q_1 = \frac{1}{3}(3+4) = 3.5$$

$F(5) = 0.008 + 0.032 + 0.142 + 0.216 + 0.240 = 0.638 < 0.75$

$F(6) = 0.008 + 0.032 + 0.142 + 0.216 + 0.240 + 0.206 = 0.844 > 0.75$

$$Q_2 = \frac{1}{2}(4+5) = 4.5$$

$$Q_3 = \frac{1}{2}(5+6) = 5.5$$

(iv) Median = $Q_2 = 4.5$

(v) Pearson's coefficient of skewness

$$S_k = \frac{\text{Mean} - \text{Median}}{\text{SD}} = \frac{4.903 - 4.5}{1.444} = 0.279$$

Example 7

Find the mean, median QD, MD, SD, β_1 and β_2 of the following probability distribution:

$X = x$	0	1	2	3	4	5	6	7	8
$p(x)$	0.004	0.036	0.1	0.232	0.280	0.204	0.112	0.028	0.004

Solution

(i) Mean = $\mu = \sum x\, p(x)$

$$= 0 + 1(0.036) + 2(0.1) + 3(0.232) + 4(0.280) + 5(0.204) + 6(0.112) + 7(0.028) + 8(0.04)$$

$$= 3.972$$

(ii) Median

$$F(3) = 0.004 + 0.036 + 0.1 + 0.232 = 0.372 < 0.5$$

$$F(4) = 0.004 + 0.036 + 0.1 + 0.232 + 0.280 = 0.652 > 0.5$$

$$\text{Median } M = \frac{1}{2}(3+4) = 3.5$$

(iii) Mode is the value of X for which $P(X = x)$ is maximum.

Mode = 4 [$\because P(X = 4) = 0.280$ is maximum probability]

(iv) Variance = $\sigma^2 = \sum x^2 p(x) - \mu^2$

$$= 0 + 1(0.036) + 4(0.1) + 9(0.232) + 16(0.28) + 25(0.204) + 36(0.112) + 49(0.028) + 64(0.004) - (3.972)^2$$

$$= 1.987$$

$$\text{SD} = \sqrt{\text{Var}(X)} = \sqrt{1.987} = 1.41$$

(v) $F(2) = 0.004 + 0.036 + 0.1 = 0.14 < 0.25$

$F(3) = 0.372 > 0.25$

$$Q_1 = \frac{1}{2}(2+3) = 2.5$$

$F(4) = 0.652 < 0.75$

$F(5) = 0.004 + 0.036 + 0.1 + 0.232 + 0.280 + 0.204 = 0.856 > 0.75$

$$Q_3 = \frac{1}{2}(4+5) = 4.5$$

$$QD = \frac{1}{2}(Q_3 - Q_1) = \frac{1}{2}(4.5-2.5) = 1$$

(vi) $MD = \sum |x-\mu| p(x)$

$= 3.972(0.004) + 2.972(0.036) + 1.972(0.1) + 0.972(0.232)$
$+ 0.028\ (0.28) + 1.028(0.204) + 2.028(0.112)$
$+ 3.028(0.028) + 4.028(0.004)$

$= 1.091$

(vii) $\mu_1' = \mu = 3.972$

$\mu_2' = E(X^2) = \sum x^2 p(x)$

$= 0 + 1(0.036) + 4(0.1) + 9(0.232) + 16(0.280)$
$+ 25(0.204) + 36(0.112) + 49(0.028) + 64(0.004)$

$= 17.764$

$\mu_3' = E(X^3) = \sum x^3 p(x)$

$= 0 + 1(0.036) + 8\ (0.1) + 27(0.232) + 64(0.280) + 125(0.204)$
$+ 216(0.112) + 343(0.028) + 512(0.004)$

$= 86.364$

$\mu_4' = E(X^4) = \sum x^4 p(x)$

$= 0 + 1(0.036) + 16(0.1) + 81(0.232) + 256(0.280)$
$+ 625(0.204) + 1296(0.112) + 2401(0.028) + 4096(0.004)$

$= 448.372$

$\mu_2 = \sigma^2 = 1.987$

$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3$

$= 86.364 - 3(17.764)\ (3.972) + 2(3.972)^3$

$= 0.019$

$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$

$= 448.372 - 4(86.364)\ (3.972) + 6(17.764)\ (3.972)^2 - 3(3.972)^4$

$= 11.053$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(0.019)^2}{(1.987)^3} = 0.00005$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{11.053}{(1.987)^2} = 2.8$$

3.7 MEASURES OF STATISTICS FOR CONTINUOUS RANDOM VARIABLES

1. Mean The mean or average value (μ) of the probability distribution of a continuous random variable X is called the expectation and is denoted by $E(X)$.

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Expectation of any function $\phi(x)$ of a continuous random variable X is given by

$$E[\phi(x)] = \int_{-\infty}^{\infty} \phi(x)\, f(x)\, dx$$

2. Median The median is the point which divides the entire distribution into two equal parts. In case of a continuous distribution, the median is the point which divides the total area into two equal parts. Thus, if a continuous random variable X is defined from a to b and M is the median,

$$\int_{a}^{M} f(x)\, dx = \int_{M}^{b} f(x)\, dx = \frac{1}{2}$$

By solving any one of this equation, the median is obtained.

3. Mode The mode is value of x for which $f(x)$ is maximum. Mode is given by

$$f'(x) = 0 \text{ and } f''(x) < 0 \text{ for } a < x < b$$

4. Geometric Mean The geometric mean of the probability distribution of a continuous random variable X is given by

$$\log G = \int_{-\infty}^{\infty} (\log x) f(x)\, dx$$

5. Harmonic Mean The harmonic mean of the probability distribution of a continuous random variable X is given by

$$\frac{1}{H} = \int_{-\infty}^{\infty} \frac{1}{x} f(x)\, dx$$

6. Quartile Deviation The r^{th} quartile of the probability distribution of a continuous random variable X and denoted by Q_r, is given by

$$\int_{-\infty}^{Q_r} f(x)\, dx = \frac{r}{4},\ r = 1, 2, 3$$

Q_1 and Q_3 are called the first (lower) and the third (upper) quartiles respectively. Q_2 is the median (middle or second quartile). Quartile deviation or semi-interquartile range of the probability distribution of a continuous random variable X is given by

$$Q = \frac{1}{2}(Q_3 - Q_1)$$

7. Mean Deviation Mean deviation of the probability distribution of a continuous random variable X is given by

$$\text{MD} = \int_{-\infty}^{\infty} |x-\mu| f(x)\,dx$$

8. Standard Deviation The standard deviation of the probability distribution of a continuous random variable X is given by

$$\text{SD} = \sqrt{\text{Var}(X)} = \sigma$$

9. Variance The variance of the probability distribution of a continuous random variable X is given by

$$\text{Var}(X) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$$

$$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

10. Moments Central moments or moments about actual mean of the probability distribution of a continuous random variable X is given by

$$\mu_r = \int_{-\infty}^{\infty} (x-\mu)^r f(x)\,dx$$

Raw moments or moments about arbitrary origin of the probability distribution of a continuous random variable X is given by

$$\mu_r' = \int_{-\infty}^{\infty} (x-a)^r f(x)\,dx$$

When $a = 0$, μ_r' is called r^{th} order simple moments.

$$\mu_r' = \int_{-\infty}^{\infty} x^r f(x)\,dx$$

11. Skewness Skewness of the probability distribution of a continuous random variable X is given by

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

12. Kurtosis Kurtosis of the probability distribution of a continuous random variable X is given by

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}$$

Note The formulae of various measures of central tendency, dispersion, skewness and kurtosis of discrete probability distribution can be easily extended to the case of continuous probability distribution by simply replacing $p(x)$ by $f(x)dx$ and the summation by integration over the specified range of the variable X.

Example 1

For the continuous random variable having pdf

$$f(x) = 4x^3 \quad 0 \le x \le 1$$
$$= 0 \quad \text{otherwise}$$

Find the mean and variance of X.

Solution

$$\text{Mean} = \mu = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$= \int_{-\infty}^{0} x f(x)\,dx + \int_{0}^{1} x f(x)\,dx + \int_{1}^{\infty} x f(x)\,dx$$

$$= 0 + \int_{0}^{1} x(4x^3)\,dx + 0$$

$$= 4\int_{0}^{1} x^4\,dx$$

$$= 4\left|\frac{x^5}{5}\right|_0^1$$

$$= 4\left(\frac{1}{5} - 0\right)$$

$$= \frac{4}{5}$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

$$= \int_{-\infty}^{0} x^2 f(x)\,dx + \int_{0}^{1} x^2 f(x)\,dx + \int_{1}^{\infty} x^2 f(x)\,dx - \mu^2$$

$$= 0 + \int_{0}^{1} x^2 (4x^3)\,dx + 0 - \left(\frac{4}{5}\right)^2$$

$$= 4\int_0^1 x^5\,dx - \frac{16}{25}$$

$$= 4\left|\frac{x^6}{6}\right|_0^1 - \frac{16}{25}$$

$$= \frac{4}{6} - \frac{16}{25}$$

$$= \frac{2}{75}$$

Example 2

For the triangular distribution

$$f(x) = x \qquad 0 < x \le 1$$
$$= 2 - x \qquad 1 \le x \le 2$$
$$= 0 \qquad \textit{otherwise}$$

Find the mean and variance.

Solution

$$\mu = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$= \int_{-\infty}^{0} x f(x)\,dx + \int_0^1 x f(x)\,dx + \int_1^2 x f(x) + \int_2^{\infty} x f(x)\,dx$$

$$= 0 + \int_0^1 x \cdot x\,dx + \int_1^2 x(2-x)\,dx + 0$$

$$= \int_0^1 x^2\,dx + \int_1^2 (2x - x^2)\,dx$$

$$= \left|\frac{x^3}{3}\right|_0^1 + \left|2\frac{x^2}{2} - \frac{x^3}{3}\right|_1^2$$

$$= \left(\frac{1}{3} - 0\right) + \left[\left(4 - \frac{8}{3}\right) - \left(1 - \frac{1}{3}\right)\right]$$

$$= \frac{1}{3} + \frac{4}{3} - \frac{2}{3}$$

$$= 1$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$$

$$= \int_{-\infty}^{0} x^2 f(x)\,dx + \int_{0}^{1} x^2 f(x)\,dx + \int_{1}^{2} x^2 f(x)\,dx + \int_{2}^{\infty} x^2 f(x)\,dx - \mu^2$$

$$= 0 + \int_{0}^{1} x^2 \cdot x\,dx + \int_{1}^{2} x^2 (2-x)\,dx + 0 - 1$$

$$= \int_{0}^{1} x^3\,dx + \int_{1}^{2} (2x^2 - x^3)\,dx - 1$$

$$= \left|\frac{x^4}{4}\right|_0^1 + \left|\frac{2x^3}{3} - \frac{x^4}{4}\right|_1^2 - 1$$

$$= \left(\frac{1}{4} - 0\right) + \left[\left(\frac{16}{3} - \frac{16}{4}\right) - \left(\frac{2}{3} - \frac{1}{4}\right)\right] - 1$$

$$= \frac{7}{6} - 1$$

$$= \frac{1}{6}$$

Example 3

If the probability density function of X is given by

$$f(x) = \begin{cases} \frac{x}{2} & 0 < x \le 1 \\ \frac{1}{2} & 1 < x \le 2 \\ \frac{3-x}{2} & 2 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

Find the expected value of $f(x) = x^2 - 5x + 3$.

Solution

$$E[E\phi(x)] = \int_{-\infty}^{\infty} \phi(x) f(x)\,dx$$

$$E(x^2 - 5x + 3) = \int_{-\infty}^{\infty} (x^2 - 5x + 3) f(x)\,dx$$

$$= \int_0^1 (x^2 - 5x + 3)\frac{x}{2}\,dx + \int_1^2 (x^2 - 5x + 3)\frac{1}{2}\,dx + \int_2^3 (x^2 - 5x + 3)\left(\frac{3-x}{2}\right)dx$$

$$= \frac{1}{2}\int_0^1 (x^3 - 5x^2 + 3x)\,dx + \frac{1}{2}\int_1^2 (x^2 - 5x + 3)\,dx + \frac{1}{2}\int_2^3 (-x^3 + 8x^2 - 18x + 9)\,dx$$

$$= \frac{1}{2}\left|\frac{x^4}{4} - \frac{5x^3}{3} + \frac{3x^2}{2}\right|_0^1 + \frac{1}{2}\left|\frac{x^3}{3} - \frac{5x^2}{2} + 3x\right|_1^2 + \frac{1}{2}\left|-\frac{x^4}{4} + \frac{8x^3}{3} - \frac{18x^2}{2} + 9x\right|_2^3$$

$$= \frac{1}{2}\left(\frac{1}{4} - \frac{5}{3} + \frac{3}{2}\right) + \frac{1}{2}\left(\frac{8}{3} - 10 + 6 - \frac{1}{3} + \frac{5}{2} - 3\right) + \frac{1}{2}\left(-\frac{81}{4} + \frac{216}{3} - \frac{162}{2} + 27 + \frac{16}{4} - \frac{64}{3} + \frac{72}{2} - 18\right)$$

$$= \frac{1}{24} - \frac{13}{12} - \frac{19}{24}$$

$$= -\frac{11}{6}$$

Example 4

A continuous random variable has the probability density function

$$f(x) = kxe^{-\lambda x} \quad x \geq 0, \lambda > 0$$
$$= 0 \quad \textit{otherwise}$$

Determine (i) k, (ii) mean, and (iii) variance.

Solution

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{0} f(x)\,dx + \int_0^{\infty} f(x)\,dx = 1$$

$$0 + \int_0^{\infty} k\,x e^{-\lambda x}\,dx = 1$$

$$k\int_0^\infty x e^{-\lambda x}\,dx = 1$$

$$k\left|x\frac{e^{-\lambda x}}{-\lambda} - 1\frac{e^{-\lambda x}}{\lambda^2}\right|_0^\infty = 1$$

$$k\left[(0-0)-\left(0-\frac{1}{\lambda^2}\right)\right] = 1$$

$$k = \lambda^2$$

Hence, $f(x) = \lambda^2 x e^{-\lambda x} \quad x \geq 0, \lambda = 0$

$= 0 \quad$ otherwise

(ii) Mean $= \mu = \int_{-\infty}^{\infty} x f(x)\,dx$

$$= \int_{-\infty}^{0} x f(x)\,dx + \int_0^\infty x f(x)\,dx$$

$$= 0 + \int_0^\infty x \lambda^2 x e^{-\lambda x}\,dx$$

$$= \lambda^2 \int_0^\infty x^2 e^{-\lambda x}\,dx$$

$$= \lambda^2 \left|x^2\left(\frac{e^{-\lambda x}}{-\lambda}\right) - 2x\left(\frac{e^{-\lambda x}}{\lambda^2}\right) + 2\left(\frac{e^{-\lambda x}}{-\lambda^3}\right)\right|_0^\infty$$

$$= \lambda^2\left[(0-0+0)-\left(0-0-\frac{2}{\lambda^3}\right)\right]$$

$$= \frac{2}{\lambda}.$$

(iii) Variance $= \sigma^2 = \int_{-\infty}^{\infty} x^2 f(x)\,dx - \mu^2$

$$= \int_{-\infty}^{0} x^2 f(x)\,dx + \int_0^\infty x^2 f(x)\,dx - \mu^2$$

$$= 0 + \int_0^\infty x^2 \lambda^2 x e^{-\lambda x}\,dx - \left(\frac{2}{\lambda}\right)^2$$

$$= \lambda^2 \int_0^\infty x^3 e^{-\lambda x}\,dx - \frac{4}{\lambda^2}$$

$$= \lambda^2 \left| x^3\left(\frac{e^{-\lambda x}}{-\lambda x}\right) - 3x^2\left(\frac{e^{-\lambda x}}{\lambda^2}\right) + 6x\left(\frac{e^{-\lambda x}}{-\lambda^3}\right) - 6\left(\frac{e^{-\lambda x}}{\lambda^4}\right)\right|_0^\infty - \frac{4}{\lambda^2}$$

$$= \lambda^2 \left[(0-0+0-0) - \left(0-0+0-\frac{6}{\lambda^4}\right)\right] - \frac{4}{\lambda^2}$$

$$= \frac{6}{\lambda^2} - \frac{4}{\lambda^2}$$

$$= \frac{2}{\lambda^2}$$

Example 5

The probability density $f(x)$ of a continuous random variable is given by $f(x) = k\,e^{-|x|}$, $-\infty < x < \infty$ (i) show that $k = \frac{1}{2}$, and (ii) find the mean and variance of the distribution. (iii) Also, find the probability that the variate lies between 0 and 4.

Solution

(i) Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{\infty} k\,e^{-|x|}\,dx = 1$$

$$k\int_{-\infty}^{\infty} e^{-|x|}\,dx = 1$$

$$2k\int_{0}^{\infty} e^{-|x|}\,dx = 1 \qquad \left[\because\ e^{-|x|} \text{ is an even function}\right]$$

$$2k\int_{0}^{\infty} e^{-x}\,dx = 1 \qquad \left[\because\ |x| = x \quad 0 \le x \le \infty\right]$$

$$2k\left|-e^{-x}\right|_0^\infty = 1$$

$$-2k\,(0-1) = 1$$

$$k = \frac{1}{2}$$

Hence, $f(x) = \frac{1}{2} e^{-|x|} \qquad -\infty < x < \infty$

(ii) $\mu = \int_{-\infty}^{\infty} x\, f(x)\, dx$

$= \frac{1}{2} \int_{-\infty}^{\infty} x e^{-|x|}\, dx$

$= 0 \qquad [\because$ the integrand is an odd function$]$

(iii) $\text{Var}(X) = \sigma^2 = \int_{-\infty}^{\infty} x^2\, f(x)\, dx - \mu^2$

$= \frac{1}{2} \int_{-\infty}^{\infty} x^2\, e^{-|x|}\, dx - 0$

$= 2\left(\frac{1}{2}\right) \int_{0}^{\infty} x^2\, e^{-|x|}\, dx$ [$\because$ the integrand is an even function]

$= \int_{0}^{\infty} x^2\, e^{-|x|}\, dx$

$= \left| x^2 \frac{e^{-x}}{-1} - 2x \frac{e^{-x}}{1} + 2 \frac{e^{-x}}{-1} \right|_0^{\infty}$

$= 0 - (-2)$

$= 2$

(iii) Probability that the variate lies between 0 and 4

$P(0 < X < 4) = \int_0^4 f(x)\, dx$

$= \frac{1}{2} \int_0^4 e^{-|x|}\, dx$

$= \frac{1}{2} \int_0^4 e^{-x}\, dx \qquad [\because \ |x| = x \quad 0 < x < 4]$

$= -\frac{1}{2} |e^{-x}|_0^4$

$= -\frac{1}{2}(e^{-4} - 1)$

$= 0.4908$

Example 6

The daily consumption of electric power is a random variable X with probability density function

$$f(x) = k\,x\,e^{-\frac{x}{3}} \qquad x > 0$$
$$= 0 \qquad x \le 0$$

Find the value of k, the expectation of X, and the probability that on a given day, the electric consumption is more than the expected value.

Solution

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{0} f(x)\,dx + \int_{0}^{\infty} f(x)\,dx = 1$$

$$0 + \int_{0}^{\infty} k\,x\,e^{-\frac{x}{3}}\,dx = 1$$

$$k\left|x\left(\frac{e^{-\frac{x}{3}}}{-\frac{1}{3}}\right) - (1)\left(\frac{e^{-\frac{x}{3}}}{\frac{1}{9}}\right)\right|_{0}^{\infty} = 1$$

$$k\left[(0-0)-(0-9)\right] = 1$$

$$9k = 1$$

$$k = \frac{1}{9}$$

Hence, $f(x) = \frac{1}{9}xe^{-\frac{x}{3}} \qquad x > 0$

$$= 0 \qquad x \le 0$$

$$E(X) = \int_{-\infty}^{\infty} x\,f(x)\,dx$$

$$= \int_{-\infty}^{0} x\,f(x)\,dx + \int_{0}^{\infty} x\,f(x)\,dx$$

$$= 0 + \int_{0}^{\infty} x \cdot \frac{1}{9}\,x\,e^{-\frac{x}{3}}\,dx$$

$$= \frac{1}{9}\int_{0}^{\infty} x^2\,e^{-\frac{x}{3}}\,dx$$

$$= \frac{1}{9}\left| x^2\left(\frac{e^{-\frac{x}{3}}}{-\frac{1}{3}}\right) - 2x\left(\frac{e^{-\frac{x}{3}}}{\frac{1}{9}}\right) + 2\left(\frac{e^{-\frac{x}{3}}}{-\frac{1}{27}}\right)\right|_0^\infty$$

$$= \frac{1}{9}(0-0+0+54)$$

$$= 6$$

$$P(X>6) = \int_0^6 f(x)\,dx$$

$$= \int_0^6 \frac{1}{9} x e^{-\frac{x}{3}}\,dx$$

$$= \frac{1}{9}\int_0^6 x e^{-\frac{x}{3}}\,dx$$

$$= \frac{1}{9}\left| x\left(\frac{e^{-\frac{x}{3}}}{-\frac{1}{3}}\right) - 1\left(\frac{e^{-\frac{x}{3}}}{\frac{1}{9}}\right)\right|_0^\infty$$

$$= \frac{1}{9}\left[(0-0) - \left(-18e^{-2} - 9e^{-2}\right)\right]$$

$$= 3e^{-2}$$

$$= 0.406$$

Example 7

Let X be a random variable with E(X) = 10 and Var(X) = 25. Find the positive values of a and b such that Y = aX – b has an expectation of 0 *and a variance of* 1.

Solution

$$E(Y) = E(aX-b)$$

$$0 = aE(X) - b$$

$$= a(10) - b$$

$$10a - b = 0$$

$$\text{Var}(Y) = \text{Var}(aX-b)$$

$$1 = a^2\ \text{Var}(X)$$

$$= a^2(25)$$

$$25a^2 = 1$$

$$a = \frac{1}{5}$$

$$b = 2$$

Example 8

A continuous random variable X is distributed over the interval [0, 1] *with pdf* $f(x) = ax^2 + bx$, *where a, b are constants. If the mean of X is* 0.5, *find the values of a and b.*

Solution

Since $f(x)$ is probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx = 1$$

$$0 + \int_{0}^{1} (ax^2 + bx)\,dx + 0 = 1$$

$$\left| \frac{ax^3}{3} + \frac{bx^2}{2} \right|_0^1 = 1$$

$$\frac{a}{3} + \frac{b}{2} = 1$$

$$2a + 3b = 6 \quad \ldots(1)$$

Also, $\mu = 0.5$

$$\int_{0}^{1} x\,f(x)\,dx = 0.5$$

$$\int_{0}^{1} x\,(ax^2 + bx)\,dx = 0.5$$

$$\int_{0}^{1} (ax^3 + bx^2)\,dx = 0.5$$

$$\left| \frac{ax^4}{4} + \frac{bx^3}{3} \right|_0^1 = 0.5$$

$$\frac{a}{4} + \frac{b}{3} = 0.5$$

$$3a + 4b = 6 \quad \ldots(2)$$

Solving Eqs (1) and (2),

$a = -6, \quad b = 6$

Example 9

A continuous random variable X has the pdf defined by $f(x) = A + Bx$, $0 \le x \le 1$. *If the mean of the distribution is* $\frac{1}{3}$, *find A and B.*

Solution

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx = 1$$

$$0 + \int_{0}^{1} (A + Bx)\,dx + 0 = 1$$

$$\left| Ax + \frac{Bx^2}{2} \right|_0^1 = 1$$

$$A + \frac{B}{2} = 1 \qquad \ldots(1)$$

Also, $\mu = \frac{1}{3}$

$$\int_{-\infty}^{\infty} x\,f(x)\,dx = \frac{1}{3}$$

$$\int_{-\infty}^{0} x\,f(x)\,dx + \int_{0}^{1} x\,f(x)\,dx = \frac{1}{3}$$

$$0 + \int_{0}^{1} x\,(A + Bx)\,dx = \frac{1}{3}$$

$$\int_{0}^{1} (Ax + Bx^2)\,dx = \frac{1}{3}$$

$$\left| \frac{Ax^2}{2} + \frac{Bx^3}{3} \right|_0^1 = \frac{1}{3}$$

$$\frac{A}{2} + \frac{B}{3} = \frac{1}{3}$$

$$3A + 2B = 2 \qquad \ldots(2)$$

Solving Eqs (1) and (2),

$A = 2, \quad B = -2$

Example 10

A continuous random variable has probability density function $f(x) = 6(x - x^2) \quad 0 \le x \le 1$.

Find the (i) mean, (ii) variance, (iii) median, and (iv) mode.

Solution

(i) $\mu = \int_{-\infty}^{\infty} x\, f(x)\, dx$

$= \int_{-\infty}^{0} x f(x)\, dx + \int_{0}^{1} x f(x)\, dx + \int_{1}^{\infty} x f(x)\, dx$

$= 0 + \int_{0}^{1} x\, 6(x - x^2)\, dx + 0$

$= 6\int_{0}^{1} (x^2 - x^3)\, dx$

$= 6\left|\frac{x^3}{3} - \frac{x^4}{4}\right|_0^1$

$= 6\left(\frac{1}{3} - \frac{1}{4}\right)$

$= \frac{1}{2}$

(ii) $\text{Var}(X) = \int_{-\infty}^{\infty} x^2 f(x)\, dx - \mu^2$

$= \int_{-\infty}^{0} x^2 f(x)\, dx + \int_{0}^{1} x^2 f(x)\, dx + \int_{1}^{\infty} x^2 f(x)\, dx - \mu^2$

$= 0 + \int_0^1 x^2\, 6(x - x^2)\, dx + 0 - \frac{1}{4}$

$= 6\int_0^1 (x^3 - x^4)\, dx - \frac{1}{4}$

$= 6\left|\frac{x^4}{4} - \frac{x^5}{5}\right|_0^1 - \frac{1}{4}$

$= 6\left(\frac{1}{4} - \frac{1}{5}\right) - \frac{1}{4}$

$$= \frac{6}{20} - \frac{1}{4}$$

$$= \frac{1}{20}$$

(iii) $$\int_a^M f(x)\,dx = \int_M^b f(x)\,dx = \frac{1}{2}$$

$$\int_0^M 6(x - x^2)\,dx = \frac{1}{2}$$

$$6\left|\frac{x^2}{2} - \frac{x^3}{3}\right|_0^M = \frac{1}{2}$$

$$6\left(\frac{M^2}{2} - \frac{M^3}{3}\right) = \frac{1}{2}$$

$$3M^2 - 2M^3 = \frac{1}{2}$$

$$4M^3 - 6M^2 + 1 = 0$$

$$(2M - 1)(2M^2 - 2M - 1) = 0$$

$$M = \frac{1}{2} \quad \text{or} \quad M = \frac{1 \pm \sqrt{3}}{2}$$

$M = \frac{1}{2}$ lies in (0, 1)

Hence, median $M = \frac{1}{2}$

(iv) Mode is the value of x for which $f(x)$ is maximum. For $f(x)$ to be maximum, $f'(x) = 0$ and $f''(x) < 0$.

$$f'(x) = 0$$

$$6(1 - 2x) = 0$$

$$x = \frac{1}{2}$$

$$f''(x) = -12x$$

At $x = \frac{1}{2}$, $f''(x) = -12 < 0$

Hence, $f(x)$ is maximum at $x = \frac{1}{2}$.

Mode $= \frac{1}{2}$

Example 11

The probability density function of a random variable X is

$$f(x) = \frac{1}{2}\sin x \quad 0 \le x \le \pi$$
$$= 0 \qquad \text{otherwise}$$

Find the mean, mode, and median of the distribution and also, find the probability between 0 and $\frac{\pi}{2}$.

Solution

(i) $\mu = \int_{-\infty}^{\infty} f(x)\,dx$

$$= \int_{-\infty}^{0} x f(x)\,dx + \int_{0}^{\pi} x f(x)\,dx + \int_{\pi}^{\infty} x f(x)\,dx$$

$$= 0 + \int_{0}^{\pi} x\left(\frac{1}{2}\sin x\right)dx + 0$$

$$= \frac{1}{2}\int_{0}^{\pi} x \sin x\,dx$$

$$= \frac{1}{2}\left|-x\cos x + \sin x\right|_0^{\pi}$$

$$= \frac{\pi}{2}$$

(ii) Mode is the value of x for which $f(x)$ is maximum. For $f(x)$ to be maximum, $f'(x) = 0$ and $f''(x) < 0$.

$$f'(x) = 0$$
$$\cos x = 0$$
$$x = \frac{\pi}{2}$$
$$f''(x) = -\frac{1}{2}\sin x$$

At $x = \frac{\pi}{2}$, $f''(x) = -\frac{1}{2} < 0$

Hence, $f(x)$ is maximum of $x = \frac{\pi}{2}$.

$$\text{Mode} = \frac{\pi}{2}$$

(iii) $$\int_a^M f(x)\,dx = \int_M^b f(x)\,dx = \frac{1}{2}$$

$$\int_0^M \frac{1}{2}\sin x\,dx = \int_M^{\pi} \frac{1}{2}\sin x\,dx = \frac{1}{2}$$

$$\int_0^M \frac{1}{2}\sin x\,dx = \frac{1}{2}$$

$$-\frac{1}{2}|\cos x|_0^M = \frac{1}{2}$$

$$-\frac{1}{2}(\cos M - 1) = \frac{1}{2}$$

$$1 - \cos M = 0$$

$$\cos M = 0$$

$$M = \frac{\pi}{2}$$

Hence, median $M = \frac{\pi}{2}$

(iv) $$P\left(0 < X < \frac{\pi}{2}\right) = \int_0^{\frac{\pi}{2}} f(x)\,dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2}\sin x\,dx$$

$$= -\frac{1}{2}|\cos x|_0^{\frac{\pi}{2}}$$

$$= -\frac{1}{2}(0-1)$$

$$= \frac{1}{2}$$

Example 12

The cumulative distribution function of a continuous random variable X is $F(x) = 1 - e^{-2x} \quad x \geq 0$

$= 0 \quad x < 0$

Find the (i) the probability density function, (ii) mean, and (iii) variance.

Solution

(i) $f(x) = \frac{d}{dx} F(x)$

$$f(x) = \frac{1}{2} e^{-2x} \quad x \geq 0$$
$$= 0 \quad x < 0$$

(ii) $\mu = \int_{-\infty}^{\infty} x\, f(x)\, dx$

$$= \int_{-\infty}^{0} x\, f(x)\, dx + \int_{0}^{\infty} x\, f(x)\, dx$$
$$= 0 + \int_{0}^{\infty} x \cdot \frac{1}{2} e^{-2x}\, dx$$
$$= \frac{1}{2} \int_{0}^{\infty} x\, e^{-2x}\, dx$$
$$= \frac{1}{2} \left| x\left(\frac{e^{-2x}}{-2}\right) - 1\left(\frac{e^{-2x}}{4}\right) \right|_{0}^{\infty}$$
$$= \frac{1}{2}\left[(0-0) - \left(0 - \frac{1}{4}\right)\right]$$
$$= \frac{1}{8}$$

(iii) $\text{Var}(X) = \int_{-\infty}^{\infty} x^2\, f(x)\, dx - \mu^2$

$$= \int_{-\infty}^{0} x^2\, f(x)\, dx + \int_{0}^{\infty} x^2\, f(x)\, dx - \mu^2$$
$$= 0 + \int_{0}^{\infty} x^2 \cdot \frac{1}{2} e^{-2x}\, dx - \left(\frac{1}{8}\right)^2$$
$$= \frac{1}{2} \int_{0}^{\infty} x^2\, e^{-2x} dx - \frac{1}{64}$$
$$= \frac{1}{2} \left| x^2\left(\frac{e^{-2x}}{-2}\right) - 2x\left(\frac{e^{-2x}}{4}\right) + 2\left(\frac{e^{-2x}}{-8}\right) \right|_{0}^{\infty} - \frac{1}{64}$$
$$= \frac{1}{2}\left[(0-0-0) - \left(0-0-\frac{1}{4}\right)\right] - \frac{1}{64}$$
$$= \frac{1}{8} - \frac{1}{64}$$
$$= \frac{7}{64}$$

Example 13

A continuous random variable X has the distribution function

$$F(x) = 0 \qquad x \le 1$$
$$= k(x-1)^4 \qquad 1 < x \le 3$$
$$= 1 \qquad x > 3$$

Determine (i) $f(x)$, (ii) k, and (iii) mean.

Solution

(i) $f(x) = \dfrac{d}{dx} F(x)$

$$f(x) = 0 \qquad x \le 1$$
$$= 4k(x-1)^3 \qquad 1 < x \le 3$$
$$= 0 \qquad x > 3$$

(ii) Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{1} f(x)\,dx + \int_{1}^{3} f(x)\,dx + \int_{3}^{\infty} f(x)\,dx = 1$$

$$0 + \int_{1}^{3} 4k(x-1)^3\,dx + 0 = 1$$

$$4k\left|\frac{(x-1)^4}{4}\right|_1^3 = 1$$

$$k(16-0) = 1$$

$$k = \frac{1}{16}$$

Hence, $f(x) = 0 \qquad x \le 1$

$$= \frac{1}{4}(x-1)^3 \qquad 1 < x \le 3$$
$$= 0 \qquad x > 3$$

(iii) $\mu = \int_{-\infty}^{\infty} x f(x)\,dx$

$$= \int_{-\infty}^{1} x f(x)\,dx + \int_{1}^{3} x f(x)\,dx + \int_{3}^{\infty} x f(x)\,dx$$

$$= 0 + \int_{1}^{3} x \cdot \frac{1}{4}(x-1)^3\,dx + 0$$

$$= \frac{1}{4}\int_{1}^{3} x(x-1)^3\,dx$$

$$= \frac{1}{4}\int_0^2 (t+1)\, t^3 \, dt \qquad \begin{bmatrix} \text{Putting} \quad x-1=t \\ \text{When } x=1,\ t=0 \\ \text{When } x=3,\ t=2 \end{bmatrix}$$

$$= \frac{1}{4}\int_0^2 (t^4 + t^3)\, dt$$

$$= \frac{1}{4}\left|\frac{t^5}{5} + \frac{t^4}{4}\right|_0^2$$

$$= \frac{1}{4}\left[\left(\frac{2^5}{5} + \frac{2^4}{4}\right) - (0)\right]$$

$$= 2.6$$

Example 14

If the density function of a random variable X is given by

$$f(x) = kx\,(1-x), \quad 0 \le x \le 1,$$

find (i) AM, (ii) HM, (iii) Median, (iv) Mode, (v) SD, (vi) MD about the mean.

Solution

(i) Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_0^1 kx(1-x)dx = 1$$

$$k\int_0^1 (x - x^2)\, dx = 1$$

$$k\left|\frac{x^2}{2} - \frac{x^3}{3}\right|_0^1 = 1$$

$$k\left(\frac{1}{2} - \frac{1}{3}\right) = 1$$

$$k = 6$$

Hence, $f(x) = 6x(1-x), \quad 0 \le x \le 1$

(ii) $\text{AM} = \mu = E(x) = \int_{-\infty}^{\infty} xf(x)\, dx$

$$= \int_0^1 x \cdot 6x(1-x)\,dx$$

$$= 6\int_0^1 (x^2 - x^3)\,dx$$

$$= 6\left|\frac{x^3}{3} - \frac{x^4}{4}\right|_0^1$$

$$= 6\left(\frac{1}{3} - \frac{1}{4}\right)$$

$$= \frac{1}{2}$$

(iii) $\frac{1}{H} = \int_{-\infty}^{\infty} \frac{1}{x} f(x)\,dx$

$$= \int_0^1 \frac{1}{x} \cdot 6x(1-x)\,dx$$

$$= 6\int_0^1 (1-x)\,dx$$

$$= 6\left|x - \frac{x^2}{2}\right|_0^1$$

$$= 6\left(1 - \frac{1}{2}\right)$$

$$= 3$$

$$H = \frac{1}{3}$$

(iv) $\int_0^M f(x)\,dx = \frac{1}{2}$

$$\int_0^M 6x(1-x)\,dx = \frac{1}{2}$$

$$6\int_0^M (x - x^2)\,dx = \frac{1}{2}$$

$$6\left|\frac{x^2}{2} - \frac{x^3}{3}\right|_0^M = \frac{1}{2}$$

$$6\left(\frac{M^2}{2}-\frac{M^3}{3}\right)=\frac{1}{2}$$

$$3M^2-2M^3=\frac{1}{2}$$

$$6M^2-4M^3=1$$

$$4M^3-6M^2+1=0$$

$$M=\frac{1}{2},\frac{1}{2}\pm\frac{\sqrt{3}}{2}$$

The values $M=\frac{1}{2}\pm\frac{\sqrt{3}}{2}$ lie outside (0, 1).

Hence, $M=\frac{1}{2}$

(v) Mode is the value of x for which $f(x)$ is maximum. For $f(x)$ to be maximum, $f'(x)=0$ and $f''(x)<0$.

$$f'(x)=0$$

$$6-12x=0$$

$$x=\frac{1}{2}$$

$$f''(x)=-12<0$$

Hence, $f(x)$ is maximum at $x=\frac{1}{2}$

Mode $=\frac{1}{2}$

As the mean, median and mode are equal, the distribution is symmetrical.

(vi) $E(X^2)=\int_{-\infty}^{\infty} x^2+f(x)\,dx$

$$=\int_0^1 x^2\cdot 6x(1-x)\,dx$$

$$=6\int_0^1 (x^3-x^4)\,dx$$

$$=6\left|\frac{x^4}{4}-\frac{x^5}{5}\right|_0^1$$

$$=6\left(\frac{1}{4}-\frac{1}{5}\right)$$

$$=\frac{3}{10}$$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2$$

$$= \frac{3}{10} - \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{20}$$

$$SD = \sqrt{\text{Var}(X)} = \sqrt{\frac{1}{20}} = \frac{1}{2\sqrt{5}}$$

(vii) Mean deviation about the mean

$$MD = \int_{-\infty}^{\infty} |x-\mu| f(x)\,dx$$

$$= \int_0^1 \left|x - \frac{1}{2}\right| 6x(1-x)\,dx$$

$$= \int_0^{\frac{1}{2}} \left(\frac{1}{2} - x\right) 6x(1-x)\,dx + \int_{\frac{1}{2}}^1 \left(x - \frac{1}{2}\right) 6x(1-x)\,dx$$

$$= \int_0^{\frac{1}{2}} (3x - 9x^2 + 6x^3)\,dx + \int_{\frac{1}{2}}^1 (-3x + 9x^2 - 6x^3)\,dx$$

$$= \left|\frac{3x^2}{2} - 3x^3 + \frac{3x^4}{2}\right|_0^{\frac{1}{2}} + \left|-\frac{3x^2}{2} + 3x^3 - \frac{3x^4}{2}\right|_{\frac{1}{2}}^1$$

$$= \left(\frac{3}{8} - \frac{3}{8} + \frac{3}{32}\right) + \left(-\frac{3}{2} + 3 - \frac{3}{2}\right) - \left(-\frac{3}{8} + \frac{3}{8} - \frac{3}{32}\right)$$

$$= \frac{3}{16}$$

Example 15

Prove that geometric mean G of the distribution

$$f(x) = 6(2-x)(x-1),\ 1 \le x \le 2$$

is given by $6 \log(16G) = 19$.

Solution

$$\log G = \int_{-\infty}^{\infty} (\log x) f(x)\,dx$$

$$= \int_{1}^{2} (\log x)\, 6(2-x)\,(x-1)\,dx$$

$$= -6\int_{1}^{2} (x^2 - 3x + 2)\log x\,dx$$

$$= -6\left[\left|\left(\frac{x^3}{3} - \frac{3x^2}{2} + 2x\right)\log x\right|_1^2 - \int_1^2 \left(\frac{x^3}{3} - \frac{3x^2}{2} + 2x\right)\frac{1}{x}dx\right]$$

$$= -6\left[\left(\frac{8}{3} - 6 + 4\right)\log 2 - \int_1^2 \left(\frac{x^2}{3} - \frac{3x}{2} + 2\right)dx\right]$$

$$= -6\left[\frac{2}{3}\log 2 - \left|\frac{x^3}{9} - \frac{3x^2}{4} + 2x\right|_1^2\right]$$

$$= -6\left[\frac{2}{3}\log 2 - \left(\frac{8}{9} - 3 + 4\right) + \left(\frac{1}{9} - \frac{3}{4} + 2\right)\right]$$

$$= -6\left[\frac{2}{3}\log 2 - \frac{17}{9} + \frac{49}{36}\right]$$

$$= -4\log 2 + \frac{19}{6}$$

$$\log G + 4\log 2 = \frac{19}{6}$$

$$\log(G \times 2^4) = \frac{19}{6}$$

$$\log(16G) = \frac{19}{6}$$

Example 16

The probability distribution of a random variable X is

$$f(x) = k \sin\frac{\pi}{5}x, \quad 0 \le x \le 5$$

Determine the constant k and obtain the median and quartiles of the distribution.

Solution

Since $f(x)$ is a probability distribution,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_0^5 k \sin\frac{\pi}{5}x\,dx = 1$$

$$k\left|\frac{-\cos\frac{\pi}{5}x}{\frac{\pi}{5}}\right|_0^5 = 1$$

$$\frac{5k}{\pi}(-\cos\pi + \cos 0) = 1$$

$$\frac{5k}{\pi}[-(-1)+1] = 1$$

$$\frac{10k}{\pi} = 1$$

$$k = \frac{\pi}{10}$$

Hence, $$f(x) = \frac{\pi}{10}\sin\frac{\pi}{5}x, \quad 0 \le x \le 5$$

The r^{th} quartile Q_r is given by

$$\int_{-\infty}^{Q_r} f(x)\,dx = \frac{r}{4}, \quad r = 1, 2, 3$$

$$\int_0^{Q_r} \frac{\pi}{10}\sin\frac{\pi}{5}x\,dx = \frac{r}{4}$$

$$\frac{\pi}{10}\left|\frac{-\cos\frac{\pi}{5}x}{\frac{\pi}{5}}\right|_0^{Q_r} = \frac{r}{4}$$

$$\frac{1}{2}\left(-\cos\frac{\pi}{5}Q_r + \cos 0\right) = \frac{r}{4}$$

$$-\cos\frac{\pi}{5}Q_r + 1 = \frac{r}{2}$$

$$\cos\frac{\pi}{5}Q_r = 1-\frac{r}{2}$$

$$\frac{\pi}{5}Q_r = \cos^{-1}\left(1-\frac{r}{2}\right)$$

$$Q_r = \frac{5}{\pi}\cos^{-1}\left(1-\frac{r}{2}\right)$$

$$Q_1 = \frac{5}{\pi}\cos^{-1}\left(1-\frac{1}{2}\right) = \frac{5}{\pi}\cos^{-1}\left(\frac{1}{2}\right) = \frac{5}{\pi}\left(\frac{\pi}{3}\right) = \frac{5}{3}$$

$$Q_2 = \frac{5}{\pi}\cos^{-1}(1-1) = \frac{5}{\pi}\cos^{-1}(0) = \frac{5}{\pi}\left(\frac{\pi}{2}\right) = \frac{5}{2}$$

$$Q_3 = \frac{5}{\pi}\cos^{-1}\left(1-\frac{3}{2}\right) = \frac{5}{\pi}\cos^{-1}\left(-\frac{1}{2}\right) = \frac{5}{\pi}\left(\frac{2\pi}{3}\right) = \frac{10}{3}$$

$$\text{Median} = Q_2 = \frac{5}{2}$$

Example 17

Find the median, mode and quartile deviation of continuous random variable X, given that its density functions is

$$f(x) = \frac{k}{1+x^2}, \quad -\infty < x < \infty.$$

Solution

(i) Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_{-\infty}^{\infty} \frac{k}{1+x^2}dx = 1$$

$$2k\int_{0}^{\infty} \frac{1}{1+x^2}dx = 1 \quad \left[\because \int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx, \text{ if } f(x) \text{ is even function}\right]$$

$$2k\left|\tan^{-1}x\right|_0^{\infty} = 1$$

$$2k(\tan^{-1}\infty - \tan^{-1}0) = 1$$

$$2k\left(\frac{\pi}{2}\right) = 1$$

$$k = \frac{1}{\pi}$$

Hence, $$f(x) = \frac{1}{\pi(1+x^2)}, -\infty < x < \infty$$

(ii) The r^{th} quartile Q_r is given by

$$\int_{-\infty}^{Q_r} f(x)\,dx = \frac{r}{4}, \quad r = 1, 2, 3$$

$$\int_{-\infty}^{Q_r} \frac{1}{\pi(1+x^2)}\,dx = \frac{r}{4}$$

$$\frac{1}{\pi}\left|\tan^{-1}x\right|_{-\infty}^{Q_r} = \frac{r}{4}$$

$$\frac{1}{\pi}\left[\tan^{-1}Q_r - \tan^{-1}(-\infty)\right] = \frac{r}{4}$$

$$\frac{1}{\pi}\left[\tan^{-1}Q_r - \left(-\frac{\pi}{2}\right)\right] = \frac{r}{4}$$

$$\tan^{-1}Q_r = \frac{\pi}{4}(r-2)$$

$$Q_r = \tan\left\{\frac{\pi}{4}(r-2)\right\}$$

$$Q_1 = \tan\left(-\frac{\pi}{4}\right) = -1$$

$$Q_2 = \tan(0) = 0$$

$$Q_3 = \tan\left(\frac{\pi}{4}\right) = 1$$

$$QD = \frac{1}{2}(Q_3 - Q_1)$$

$$= \frac{1}{2}[1-(-1)]$$

$$= 1$$

(iii) Median $Q_2 = 0$

(iv) Mode is the value of x for which $f(x)$ is maximum. For $f(x)$ to be maximum $f'(x) = 0$ and $f''(x) < 0$.

$$f'(x)=0$$

$$-\frac{2x}{\pi(1+x^2)^2}=0$$

$$x=0$$

$$f''(x)=-\frac{2}{\pi}\left[\frac{(1+x^2)^2-x\cdot 2(1+x^2)\cdot 2x}{(1+x^2)^4}\right]$$

$$=\frac{2}{\pi}\left[\frac{3x^2-1}{(1+x^2)^3}\right]$$

$$f''(0)=-\frac{2}{\pi}<0$$

Hence, $f(x)$ is maximum at $x=0$.
Mode = 0

Example 18

Find the mean, variance and the coefficients β_1, β_2 of the distribution $f(x)=kx^2e^{-x}$, $0<x<\infty$

Solution

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx=1$$

$$\int_0^{\infty} kx^2e^{-x}\,dx=1$$

$$k\left|x^2(-e^{-x})-2x\,e^{-x}+2(-e^{-x})\right|_0^{\infty}=1$$

$$k(2e^0)=1$$

$$k=\frac{1}{2}$$

Hence,

$$f(x)=\frac{1}{2}x^2e^{-x},\ 0<x<\infty$$

$$\mu_2'=\int_{-\infty}^{\infty} x^r f(x)\,dx$$

$$=\int_0^{\infty} x^r\,\frac{1}{2}x^2e^{-x}\,dx$$

$$= \frac{1}{2}\int_0^\infty e^{-x}x^{r+2}dx$$

$$= \frac{1}{2}\lfloor r+3$$

$$= \frac{1}{2}(r+2)!$$

$$\mu_1' = \frac{1}{2}(3!) = 3$$

$$\mu_2' = \frac{1}{2}(4!) = 12$$

$$\mu_3' = \frac{1}{2}(5!) = 60$$

$$\mu_4' = \frac{1}{2}(6!) = 360$$

$$\mu_2 = \mu_2' - (\mu_1')^2 = 12 - (3)^2 = 3$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3 = 60 - 3(12)(3) + 2(3)^3 = 6$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$$

$$= 360 - 4(6)(3) + 6(12)(3)^2 - 3(3)^4$$

$$= 45$$

$$\text{Mean} = \mu_1' = 3$$

$$\text{Variance} = \mu_2 = 3$$

$$\beta_1 = \frac{\mu_3^2}{\mu_3^2} = \frac{(6)^2}{(3)^3} = \frac{4}{3}$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{45}{(3)^2} = 5$$

Example 19

The probability density function of a random variable X is given by $f(x) = kx(2-x)$, $0 \le x \le 2$. *Find mean, variance* β_1 *and* β_2.

Solution

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_0^2 k\,x(2-x)\,dx = 1$$

$$k\int_0^2 (2x - x^2)\,dx = 1$$

$$k\left|x^2 - \frac{x^3}{3}\right|_0^2 = 1$$

$$k\left(4 - \frac{8}{3}\right) = 1$$

$$k = \frac{3}{4}$$

Hence, $f(x) = \frac{3}{4}x(2-x),\ 0 \le x \le 2$

$$\mu_r' = \int_{-\infty}^{\infty} x^r f(x)\,dx$$

$$= \int_0^2 x^r \frac{3}{4}x(2-x)dx$$

$$= \frac{3}{4}\int_0^2 x^{r+1}(2-x)dx$$

$$= \frac{2(2^{r+1})}{(r+2)(r+3)}$$

$$\mu_1' = \frac{3(2^2)}{(3)(4)} = 1$$

$$\mu_2' = \frac{3(2^3)}{(4)(5)} = \frac{6}{5}$$

$$\mu_3' = \frac{3(2^4)}{(5)(6)} = \frac{8}{5}$$

$$\mu_4' = \frac{3(2^5)}{(6)(7)} = \frac{16}{7}$$

$$\mu_2 = \mu_2' - (\mu_1')^2 = \frac{6}{5} - 1 = \frac{1}{5}$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3 = \frac{8}{5} - 3\left(\frac{6}{5}\right)(1) + 2 = 0$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$$

$$= \frac{16}{7} - 4\left(\frac{8}{5}\right)(1) + 6\left(\frac{6}{5}\right)(1)^2 - 3(1)^4$$

$$= \frac{3}{35}$$

$$\text{Mean} = \mu_1' = 1$$

$$\text{Variance} = \mu_2 = \frac{1}{5}$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\frac{3}{35}}{\left(\frac{1}{5}\right)^2} = \frac{15}{7}$$

Example 20

Show that for the symmetrical distribution

$$f(x) = \frac{2a}{\pi}\left(\frac{1}{a^2 + x^2}\right), \qquad -a \le x \le a$$

$$\mu_2 = \frac{a^2(4-\pi)}{\pi} \text{ and } \mu_4 = a^4\left(1 - \frac{8}{3\pi}\right)$$

Solution

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-a}^{a} \frac{2a}{\pi}\left(\frac{1}{a^2 + x^2}\right)dx$$

$$= \frac{2a}{\pi}\left|\frac{1}{a}\tan^{-1}\frac{x}{a}\right|_{-a}^{a}$$

$$= \frac{2}{\pi}\left|\tan^{-1}\frac{x}{a}\right|_{-a}^{a}$$

$$= \frac{2}{\pi}\left[\tan^{-1}(1) - \tan^{-1}(-1)\right]$$

$$= 1$$

Hence, $f(x)$ represents a probability density function.

$$\mu_1' = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

$$= \int_{-a}^{a} x \frac{2a}{\pi}\left(\frac{1}{a^2+x^2}\right) dx$$

$$= \frac{2a}{\pi}\int_{-a}^{a} \frac{x}{a^2+x^2} dx$$

$$= \frac{2a}{\pi}\left|\frac{1}{2}\log(a^2+x^2)\right|_{-a}^{a}$$

$$= 0 \qquad [\because \text{integrand is an odd function of } x]$$

$$u_2' = \int_{-\infty}^{\infty} x^2\, f(x)\, dx$$

$$= \int_{-a}^{a} x^2 \frac{2a}{\pi}\left(\frac{1}{a^2+x^2}\right) dx$$

$$= \frac{2a}{\pi}\int_{-a}^{a} \frac{x^2}{a^2+x^2} dx$$

$$= \frac{4a}{\pi}\int_{0}^{a} \frac{x^2+a^2-a^2}{a^2+x^2} dx$$

$$= \frac{4a}{\pi}\int_{0}^{a}\left(1-\frac{a^2}{a^2+x^2}\right) dx$$

$$= \frac{4a}{\pi}\left|x - a\tan^{-1}\frac{x}{a}\right|_0^a$$

$$= \frac{4a}{\pi}(a - a\tan^{-1} 1)$$

$$= \frac{4a}{\pi}\left(a - a\frac{\pi}{4}\right)$$

$$= \frac{a^2(4-\pi)}{\pi}$$

$$\mu_2 = \mu_2' - (\mu_1')^2 = \frac{a^2(4-\pi)}{\pi} - 0 = \frac{a^2(4-\pi)}{\pi}$$

$$\mu_4 = \mu_4' \qquad (\because \mu_1' = 0)$$

$$\mu_4 = \int_{-\infty}^{\infty} x^4 \cdot f(x)\, dx$$

$$= \int_{-a}^{a} x^4 \cdot \frac{2a}{\pi}\left(\frac{1}{a^2+x^2}\right)dx$$

$$= \frac{2a}{\pi}\int_{-a}^{a} \frac{x^4}{a^2+x^2}dx$$

$$= \frac{4a}{\pi}\int_{0}^{a}\left(x^2 - a^2 + \frac{a^4}{a^2+x^2}\right)dx$$

$$= \frac{4a}{\pi}\left|\frac{1}{3}x^3 - a^2x + a^3\tan^{-1}\frac{x}{a}\right|_0^a$$

$$= \frac{4a}{\pi}\left(\frac{a^3}{3} - a^3 + a^3\tan^{-1}1\right)$$

$$= \frac{4a}{\pi}\left(\frac{a^3}{3} - a^3 + a^3\frac{\pi}{4}\right)$$

$$= a^4\left(1 - \frac{8}{3\pi}\right)$$